

Patient 12 - Pixel Filtering Comparison

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Setup

```
library(ggplot2)
library(ggribes)
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(reshape2)

set.seed(42)
```

Load Data

```
data_dir_raw <- "/restricted/projectnb/camplab/projects/20220504_Suzuki_LN/Data/Imaging/IMC/Tumor/Parde
data_dir_filtered <- "/restricted/projectnb/camplab/home/vgupta7/tumor_imc/patient12_steinbock/raw_filt
```

List files

```
txt_files_raw      <- list.files(data_dir_raw, pattern = "\\..txt$", full.names = TRUE)
txt_files_filtered <- list.files(data_dir_filtered, pattern = "\\..txt$", full.names = TRUE)

cat("Raw files found:", length(txt_files_raw), "\n")

## Raw files found: 4
cat("Filtered files found:", length(txt_files_filtered), "\n")

## Filtered files found: 4
```

Set marker columns (skip first 6 metadata columns)

```
marker_cols <- colnames(read.delim(txt_files_raw[1], check.names = FALSE))[7:47]
```

Load, sample, and reshape a set of ROI txt files

```
load_rois <- function(txt_files, label, n_sample = 50000) {  
  all_rois <- lapply(seq_along(txt_files), function(i) {  
    df <- read.delim(txt_files[i], check.names = FALSE)  
    df_markers <- df[, marker_cols]  
    df_sample <- df_markers[sample(nrow(df_markers), n_sample), ]  
    df_sample$ROI <- paste0("ROI_", i)  
    df_sample  
  })  
  pixels <- do.call(rbind, all_rois)  
  pixels_long <- melt(pixels, id.vars = "ROI",  
                     variable.name = "marker",  
                     value.name = "intensity")  
  pixels_long$dataset <- label  
  pixels_long  
}
```

Load both raw and filtered datasets

```
pixels_raw <- load_rois(txt_files_raw, label = "Raw")  
pixels_filtered <- load_rois(txt_files_filtered, label = "Filtered")
```

Add asinh transformation to both

```
pixels_raw <- pixels_raw %>% mutate(intensity_asinh = asinh(intensity / 1))  
pixels_filtered <- pixels_filtered %>% mutate(intensity_asinh = asinh(intensity / 1))
```

Combined dataset for before/after comparison plots

```
pixels_combined <- bind_rows(pixels_raw, pixels_filtered) %>%  
  mutate(dataset = factor(dataset, levels = c("Raw", "Filtered")))
```

```
cat("Raw pixels loaded:      ", nrow(pixels_raw), "\n")
```

```
## Raw pixels loaded:      8200000
```

```
cat("Filtered pixels loaded:", nrow(pixels_filtered), "\n")
```

```
## Filtered pixels loaded: 8200000
```

Section 1: Raw Pixels, No Transformation

Histograms (ROI 1)

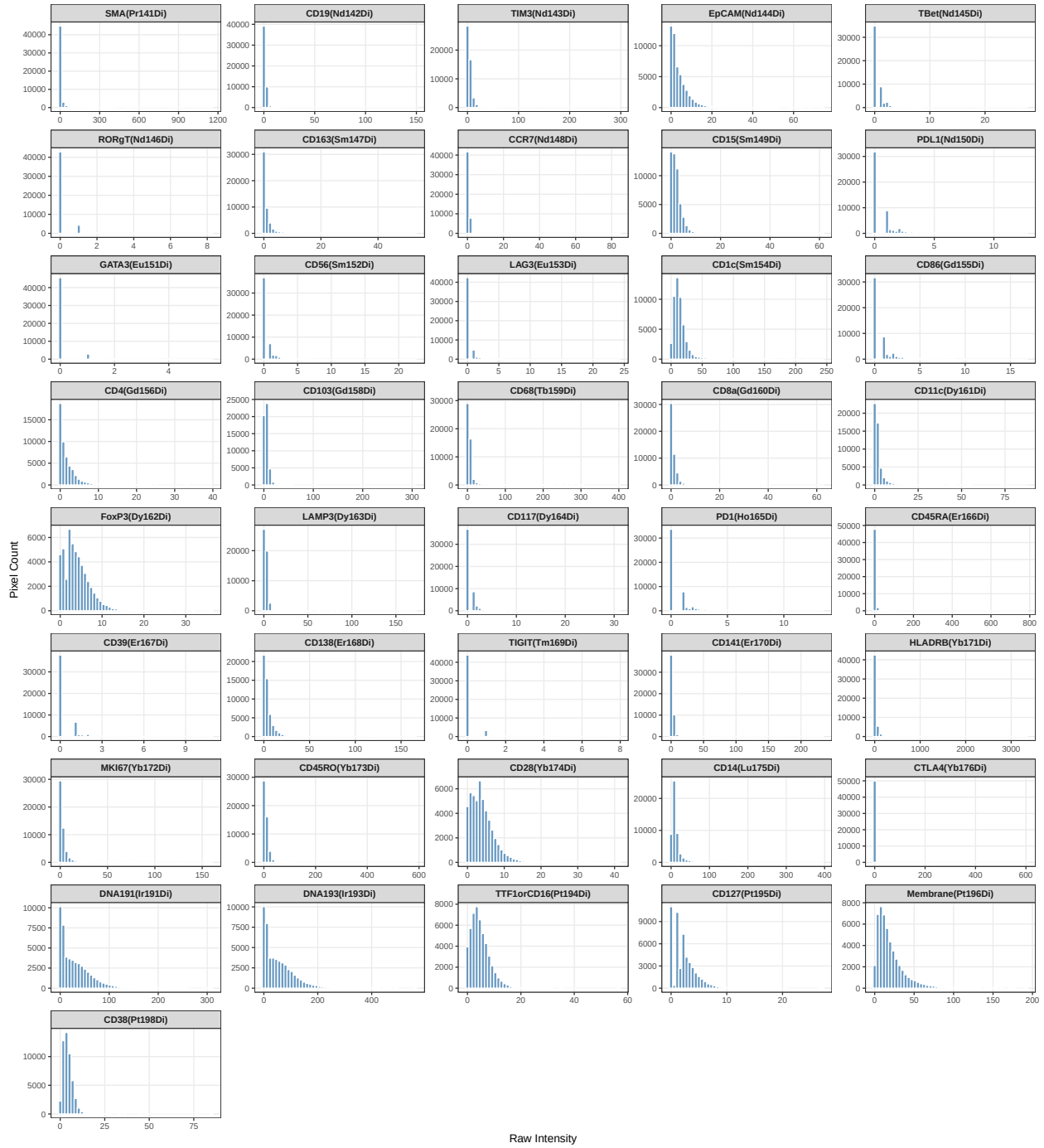
```
roi1_raw <- pixels_raw %>% filter(ROI == "ROI_1")

ggplot(roi1_raw, aes(x = intensity)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white", size = 0.1, alpha = 0.8) +
  facet_wrap(~ marker, scales = "free", ncol = 5) +
  labs(
    title = "Pixel Intensity Landscape: Patient 12 (ROI 1)",
    subtitle = "Raw values, no transformation",
    x = "Raw Intensity",
    y = "Pixel Count"
  ) +
  theme_bw(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 8),
    axis.text.y = element_text(size = 8)
  )

## Warning in geom_histogram(bins = 50, fill = "steelblue", color = "white", :
## Ignoring unknown parameters: `size`
```

Pixel Intensity Landscape: Patient 12 (ROI 1)

Raw values, no transformation



Density Ridge Plot (All ROIs)

```
ggplot(pixels_raw, aes(x = intensity, y = ROI, fill = ROI)) +
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01, show.legend = FALSE) +
  facet_wrap(~ marker, scales = "free_x", ncol = 5) +
  labs(
    title = "Patient 12 - Pixel Intensity Distributions per Marker",
    subtitle = "Raw values, no transformation",
  )
```

```

    x      = "Raw Intensity",
    y      = "Region of Interest (ROI)"
) +
theme_ridges(font_size = 10) +
theme(
  strip.text    = element_text(face = "bold", size = 9),
  axis.text.y   = element_text(size = 8),
  axis.title    = element_text(size = 11, face = "bold"),
  axis.title.x  = element_text(hjust = 0.5),
  axis.title.y  = element_text(hjust = 0.5, angle = 90)
)

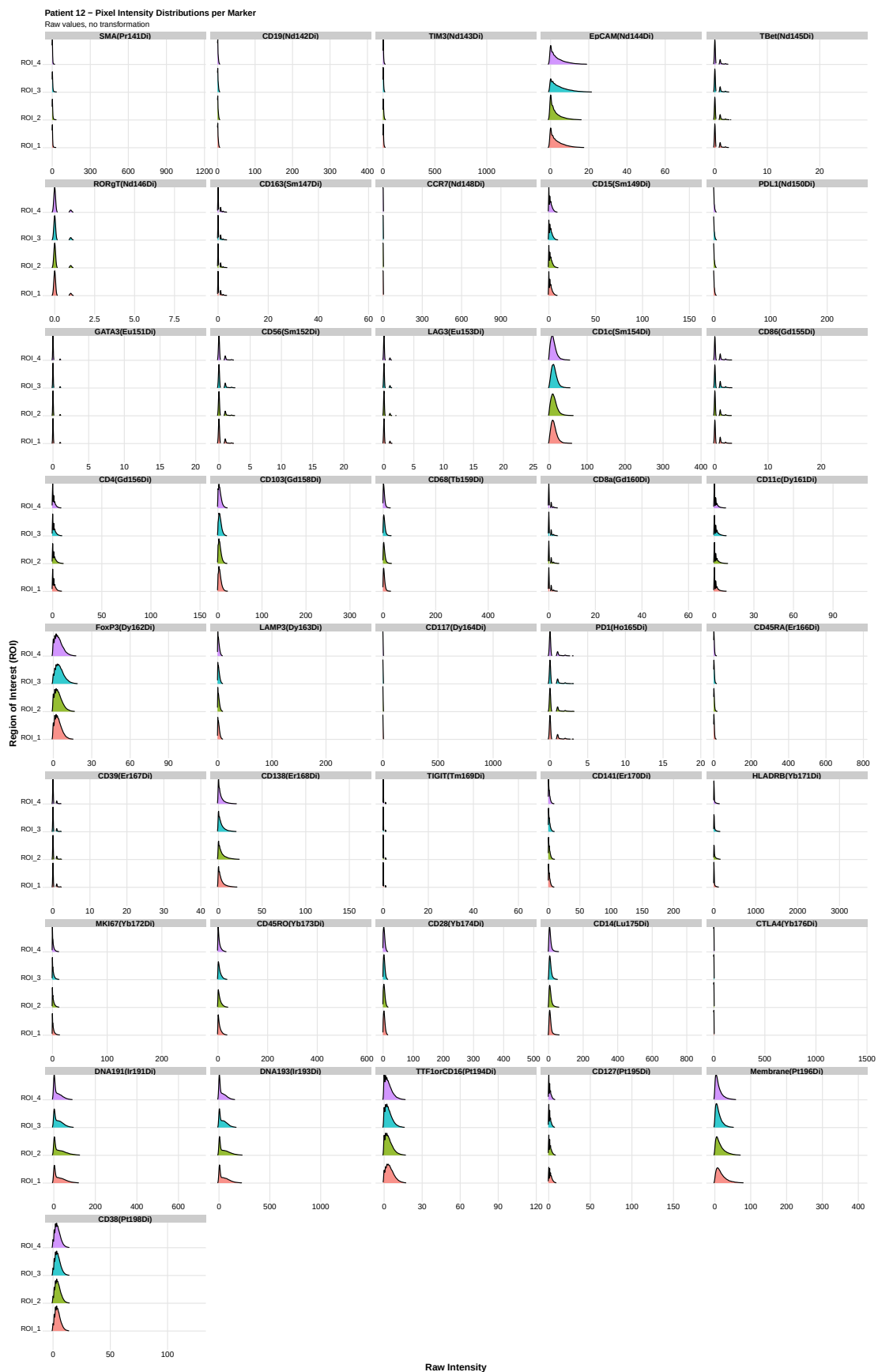
```

```

## Picking joint bandwidth of 0.278
## Picking joint bandwidth of 0.1
## Picking joint bandwidth of 0.317
## Picking joint bandwidth of 0.408
## Picking joint bandwidth of 0.0772
## Picking joint bandwidth of 0.0539
## Picking joint bandwidth of 0.0789
## Picking joint bandwidth of 0.184
## Picking joint bandwidth of 0.216
## Picking joint bandwidth of 0.0772
## Picking joint bandwidth of 0.0409
## Picking joint bandwidth of 0.0785
## Picking joint bandwidth of 0.0608
## Picking joint bandwidth of 0.801
## Picking joint bandwidth of 0.0772
## Picking joint bandwidth of 0.194
## Picking joint bandwidth of 0.36
## Picking joint bandwidth of 0.405
## Picking joint bandwidth of 0.0821
## Picking joint bandwidth of 0.181
## Picking joint bandwidth of 0.318
## Picking joint bandwidth of 0.219
## Picking joint bandwidth of 0.079
## Picking joint bandwidth of 0.0772
## Picking joint bandwidth of 0.241
## Picking joint bandwidth of 0.0786
## Picking joint bandwidth of 0.367
## Picking joint bandwidth of 0.0558

```

Picking joint bandwidth of 0.181
Picking joint bandwidth of 2
Picking joint bandwidth of 0.236
Picking joint bandwidth of 0.68
Picking joint bandwidth of 0.292
Picking joint bandwidth of 0.701
Picking joint bandwidth of 0.255
Picking joint bandwidth of 2.72
Picking joint bandwidth of 4.9
Picking joint bandwidth of 0.332
Picking joint bandwidth of 0.183
Picking joint bandwidth of 1.12
Picking joint bandwidth of 0.269



any transformation, the raw histograms look very squished with a huge spike near zero and a long right tail.

Focused Marker List (for Section 2)

```
focused_markers <- c(
  "TIM3(Nd143Di)",
  "EpCAM(Nd144Di)",
  "SMA(Pr141Di)",
  "CD19(Nd142Di)",
  "CD15(Sm149Di)",
  "CD1c(Sm154Di)",
  "CD103(Gd158Di)",
  "CD68(Tb159Di)",
  "FoxP3(Dy162Di)",
  "LAMP3(Dy163Di)",
  "HLADRB(Yb171Di)",
  "CD28(Yb174Di)",
  "CD14(Lu175Di)",
  "CD38(Pt198Di)",
  "TTF1orCD16(Pt194Di)",
  "CD163(Sm147Di)"
)
```

```
pixels_raw_focused <- pixels_raw %>% filter(marker %in% focused_markers)
cat("Focused markers confirmed:", length(unique(pixels_raw_focused$marker)), "\n")
```

```
## Focused markers confirmed: 16
```

Section 2: Raw Pixels, asinh Transformed

I did this with all 41 markers previously in p12_imc_pixel_dist.Rmd, so I'm just going to plot 16 of them here.

Histograms (ROI 1)

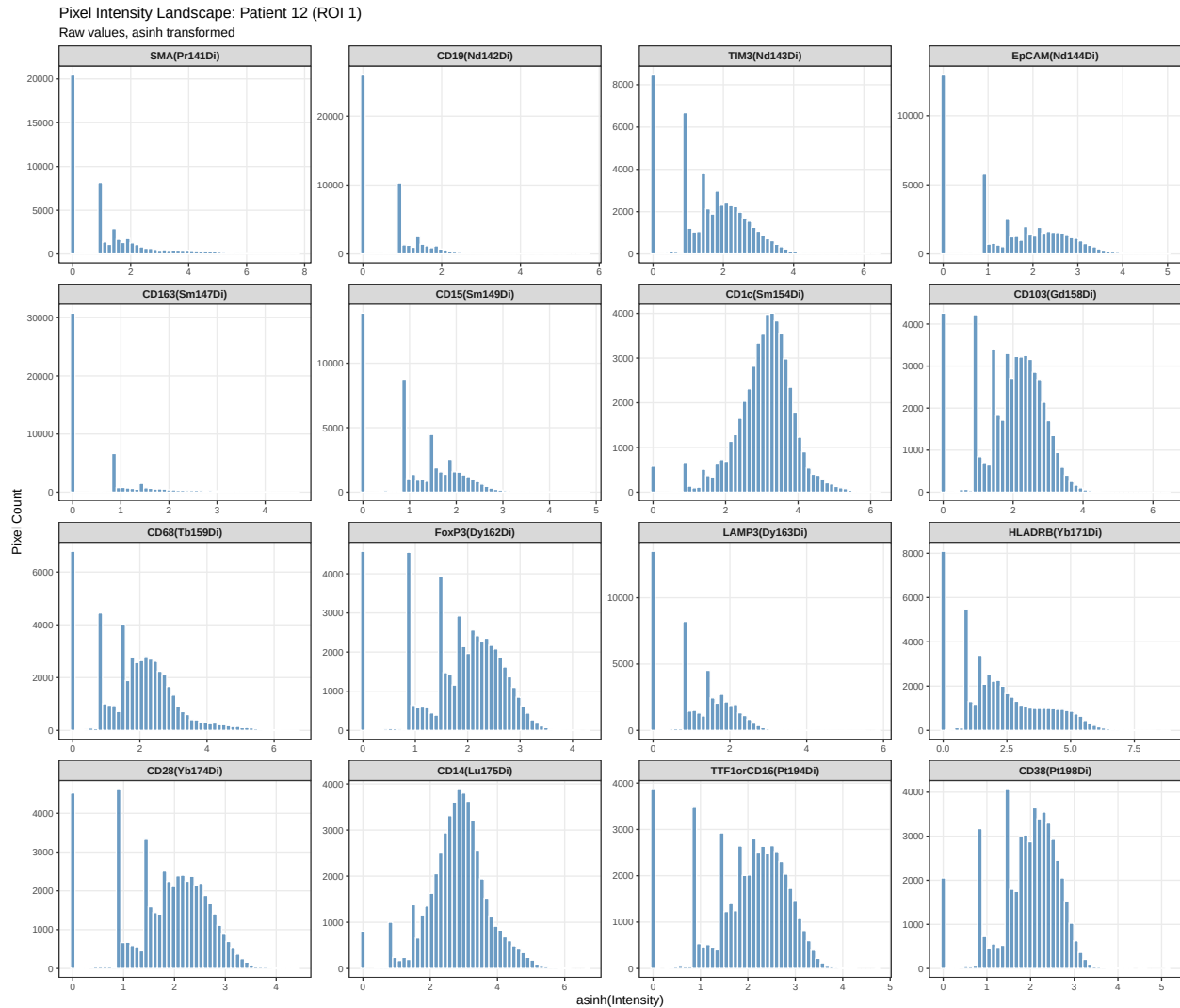
```
roi1_raw_focused <- pixels_raw_focused %>% filter(ROI == "ROI_1")

ggplot(roi1_raw_focused, aes(x = intensity_asinh)) +
  geom_histogram(bins = 50, fill = "steelblue", color = "white", size = 0.1, alpha = 0.8) +
  facet_wrap(~ marker, scales = "free", ncol = 4) +
  labs(
    title = "Pixel Intensity Landscape: Patient 12 (ROI 1)",
    subtitle = "Raw values, asinh transformed",
    x = "asinh(Intensity)",
    y = "Pixel Count"
  ) +
  theme_bw(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 8),
```



```
axis.text.y      = element_text(size = 8)
)
```

```
## Warning in geom_histogram(bins = 50, fill = "steelblue", color = "white", :
## Ignoring unknown parameters: `size`
```

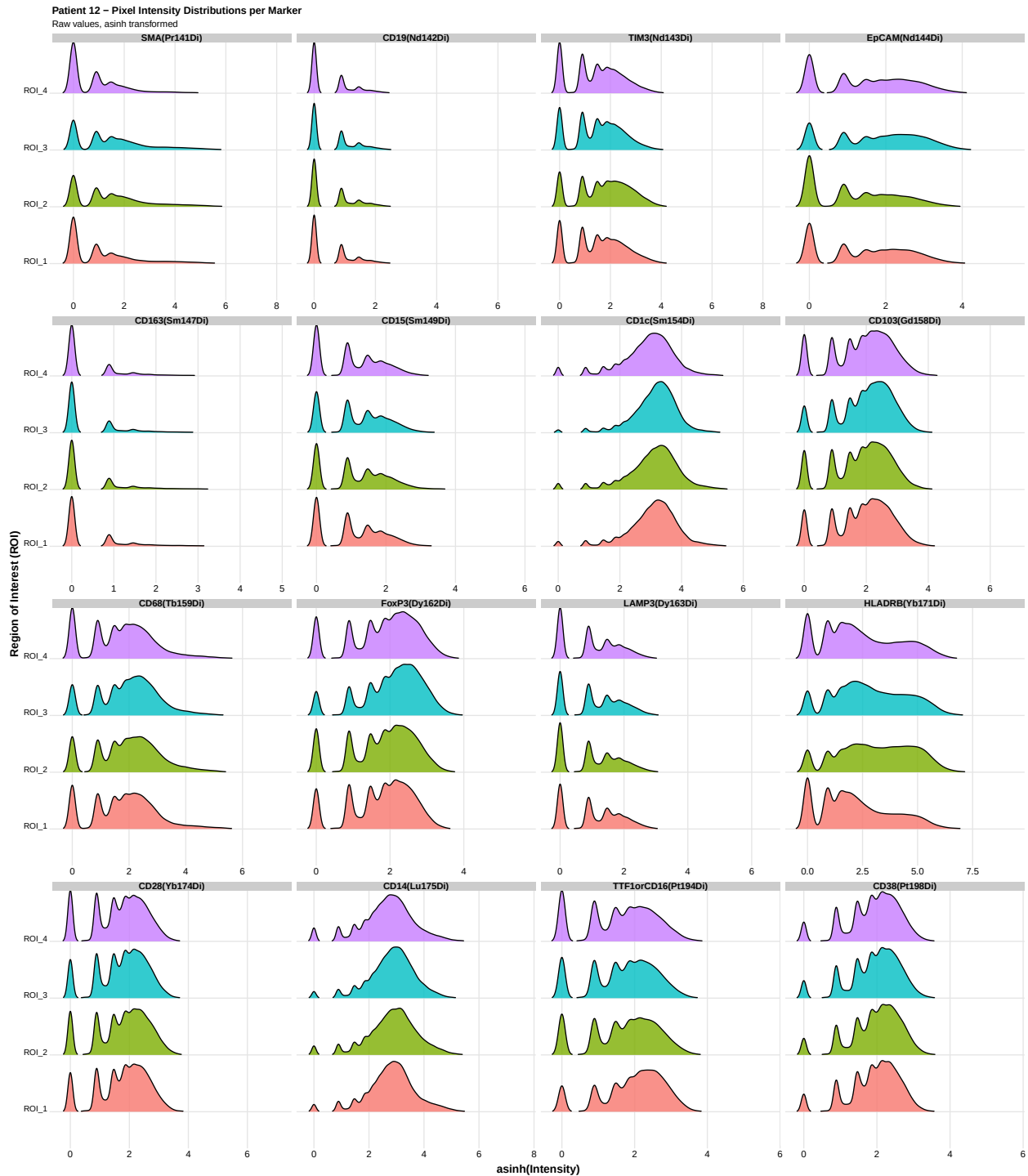


Density Ridge Plot (All ROIs)

```
ggplot(pixels_raw_focused, aes(x = intensity_asinh, y = ROI, fill = ROI)) +
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01, show.legend = FALSE) +
  facet_wrap(~ marker, scales = "free_x", ncol = 4) +
  labs(
    title = "Patient 12 - Pixel Intensity Distributions per Marker",
    subtitle = "Raw values, asinh transformed",
    x = "asinh(Intensity)",
    y = "Region of Interest (ROI)"
  ) +
  theme_ridges(font_size = 10) +
  theme(
```

```
strip.text = element_text(face = "bold", size = 9),
axis.text.y = element_text(size = 8),
axis.title = element_text(size = 11, face = "bold"),
axis.title.x = element_text(hjust = 0.5),
axis.title.y = element_text(hjust = 0.5, angle = 90)
)
```

```
## Picking joint bandwidth of 0.136
## Picking joint bandwidth of 0.0737
## Picking joint bandwidth of 0.103
## Picking joint bandwidth of 0.115
## Picking joint bandwidth of 0.0692
## Picking joint bandwidth of 0.0874
## Picking joint bandwidth of 0.0697
## Picking joint bandwidth of 0.0876
## Picking joint bandwidth of 0.109
## Picking joint bandwidth of 0.0801
## Picking joint bandwidth of 0.0872
## Picking joint bandwidth of 0.172
## Picking joint bandwidth of 0.0829
## Picking joint bandwidth of 0.0795
## Picking joint bandwidth of 0.0927
## Picking joint bandwidth of 0.0698
```



Section 3: Filtered Pixels, No Transformation

Histograms (ROI 1)

```
roi1_filtered <- pixels_filtered %>% filter(ROI == "ROI_1")

ggplot(roi1_filtered, aes(x = intensity)) +
```

```

geom_histogram(bins = 50, fill = "darkorange", color = "white", size = 0.1, alpha = 0.8) +
facet_wrap(~ marker, scales = "free", ncol = 5) +
labs(
  title      = "Pixel Intensity Landscape: Patient 12 (ROI 1)",
  subtitle   = "Filtered values, no transformation",
  x          = "Raw Intensity",
  y          = "Pixel Count"
) +
theme_bw(base_size = 11) +
theme(
  strip.text      = element_text(face = "bold", size = 9),
  panel.grid.minor = element_blank(),
  axis.text.x     = element_text(size = 8),
  axis.text.y     = element_text(size = 8)
)

```

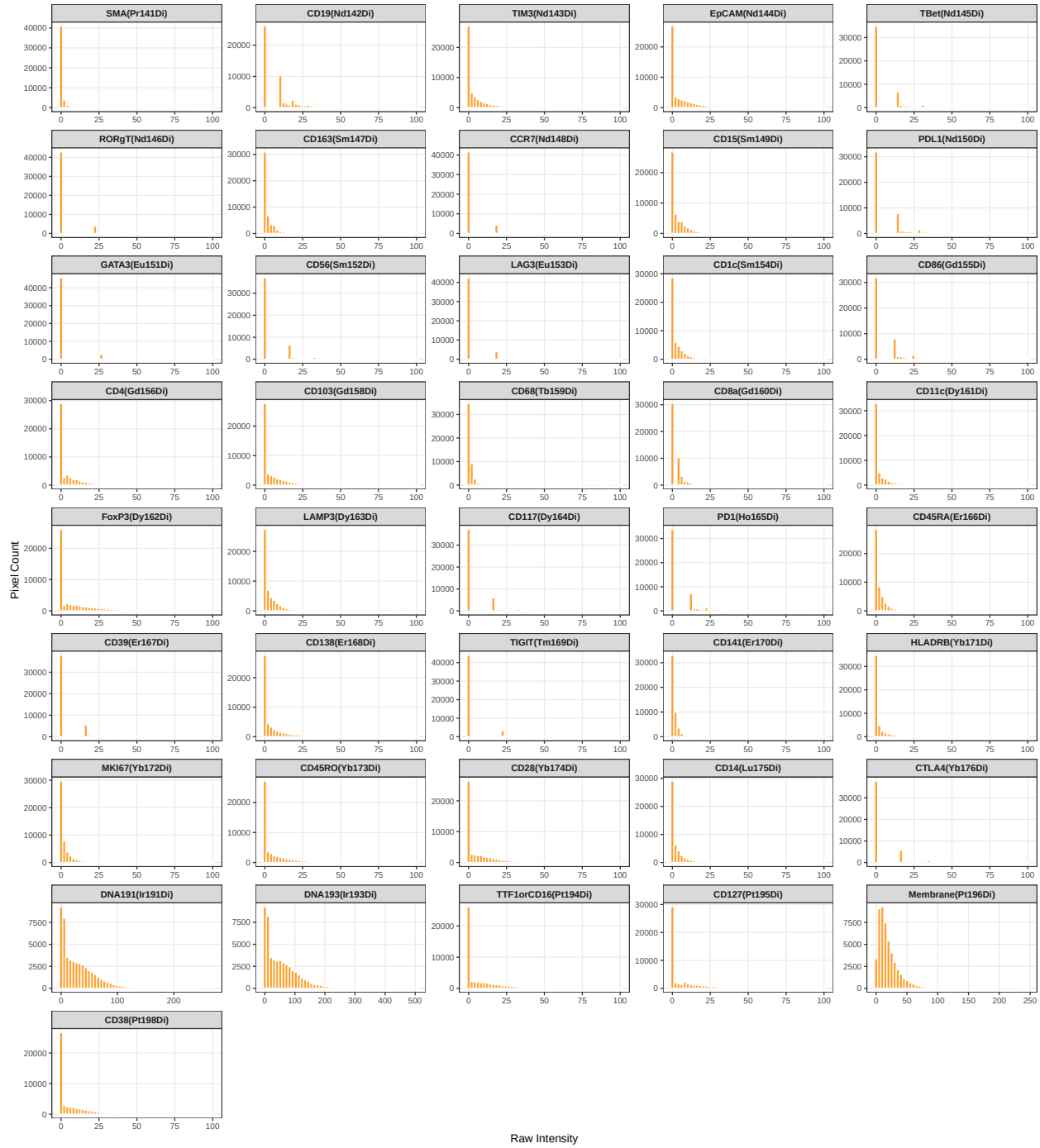
```

## Warning in geom_histogram(bins = 50, fill = "darkorange", color = "white", :
## Ignoring unknown parameters: `size`

```

Pixel Intensity Landscape: Patient 12 (ROI 1)

Filtered values, no transformation



Density Ridge Plot (All ROIs)

```
ggplot(pixels_filtered, aes(x = intensity, y = ROI, fill = ROI)) +
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01, show.legend = FALSE) +
  facet_wrap(~ marker, scales = "free_x", ncol = 5) +
  labs(
    title = "Patient 12 - Pixel Intensity Distributions per Marker",
    subtitle = "Filtered values, no transformation",
  )
```

```

    x      = "Raw Intensity",
    y      = "Region of Interest (ROI)"
) +
theme_ridges(font_size = 10) +
theme(
  strip.text    = element_text(face = "bold", size = 9),
  axis.text.y   = element_text(size = 8),
  axis.title    = element_text(size = 11, face = "bold"),
  axis.title.x  = element_text(hjust = 0.5),
  axis.title.y  = element_text(hjust = 0.5, angle = 90)
)

```

```

## Picking joint bandwidth of 0.0543
## Picking joint bandwidth of 0.918
## Picking joint bandwidth of 0.562
## Picking joint bandwidth of 0.651
## Picking joint bandwidth of 1.11
## Picking joint bandwidth of 1.22
## Picking joint bandwidth of 0.247
## Picking joint bandwidth of 1.13
## Picking joint bandwidth of 0.367
## Picking joint bandwidth of 1.07
## Picking joint bandwidth of 1.06
## Picking joint bandwidth of 1.13
## Picking joint bandwidth of 1.05
## Picking joint bandwidth of 0.364
## Picking joint bandwidth of 0.976
## Picking joint bandwidth of 0.552
## Picking joint bandwidth of 0.535
## Picking joint bandwidth of 0.136
## Picking joint bandwidth of 0.276
## Picking joint bandwidth of 0.296
## Picking joint bandwidth of 0.918
## Picking joint bandwidth of 0.415
## Picking joint bandwidth of 1.05
## Picking joint bandwidth of 0.845
## Picking joint bandwidth of 0.265
## Picking joint bandwidth of 1.3
## Picking joint bandwidth of 0.514
## Picking joint bandwidth of 1.13

```

Picking joint bandwidth of 0.125
Picking joint bandwidth of 0.305
Picking joint bandwidth of 0.207
Picking joint bandwidth of 0.648
Picking joint bandwidth of 0.872
Picking joint bandwidth of 0.37
Picking joint bandwidth of 1
Picking joint bandwidth of 2.72
Picking joint bandwidth of 4.87
Picking joint bandwidth of 0.879
Picking joint bandwidth of 0.982
Picking joint bandwidth of 1.13
Picking joint bandwidth of 0.806



ing, we still have the same number of pixels, but the hot pixels that were previously far out in the right tail have now been moved to zero. This is why the spike at zero gets bigger after filtering. However, the right tail became longer, which is unusual and unexpected. The tails should have gotten shorter because that means the outliers were removed, but now it's not certain whether that was done.

Compare max raw intensity before and after filtering

```
pixels_raw %>%
  group_by(marker) %>%
  summarise(max_raw = max(intensity)) %>%
  left_join(
    pixels_filtered %>%
      group_by(marker) %>%
      summarise(max_filtered = max(intensity)),
    by = "marker"
  ) %>%
  mutate(diff = max_filtered - max_raw) %>%
  arrange(desc(diff))
```

```
## # A tibble: 41 x 4
##   marker      max_raw max_filtered diff
##   <fct>      <dbl>      <dbl> <dbl>
## 1 RORgT(Nd146Di)    8.84         100  91.2
## 2 PD1(Ho165Di)    19.0         100  81.0
## 3 GATA3(Eu151Di)   20.3         100  79.7
## 4 CD56(Sm152Di)    23.0         100  77.0
## 5 LAG3(Eu153Di)    24.1         100  75.9
## 6 CD86(Gd155Di)    27.1         100  72.9
## 7 Tbet(Nd145Di)    27.4         100  72.6
## 8 CD39(Er167Di)    39.2         100  60.8
## 9 CD163(Sm147Di)   57.9         100  42.1
## 10 CD8a(Gd160Di)   62.4         100  37.6
## # i 31 more rows
```

Section 4: Filtered Pixels, asinh Transformed

Histograms (ROI 1)

```
ggplot(roi1_filtered, aes(x = intensity_asinh)) +
  geom_histogram(bins = 50, fill = "darkorange", color = "white", size = 0.1, alpha = 0.8) +
  facet_wrap(~ marker, scales = "free", ncol = 5) +
  labs(
    title = "Pixel Intensity Landscape: Patient 12 (ROI 1)",
    subtitle = "Filtered values, asinh transformed",
    x = "asinh(Intensity)",
    y = "Pixel Count"
  ) +
  theme_bw(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 8),
```

```
axis.text.y = element_text(size = 8)
)
```

```
## Warning in geom_histogram(bins = 50, fill = "darkorange", color = "white", :
## Ignoring unknown parameters: `size`
```



Density Ridge Plot (All ROIs)

```
ggplot(pixels_filtered, aes(x = intensity_asinh, y = ROI, fill = ROI)) +  
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01, show.legend = FALSE) +  
  facet_wrap(~ marker, scales = "free_x", ncol = 5) +  
  labs(  
    title = "Patient 12 - Pixel Intensity Distributions per Marker",  
    subtitle = "Filtered values, asinh transformed",  
    x = "asinh(Intensity)",  
    y = "Region of Interest (ROI)"  
  ) +  
  theme_ridges(font_size = 10) +  
  theme(  
    strip.text = element_text(face = "bold", size = 9),  
    axis.text.y = element_text(size = 8),  
    axis.title = element_text(size = 11, face = "bold"),  
    axis.title.x = element_text(hjust = 0.5),  
    axis.title.y = element_text(hjust = 0.5, angle = 90)  
  )
```

Picking joint bandwidth of 0.0505

Picking joint bandwidth of 0.179

Picking joint bandwidth of 0.156

Picking joint bandwidth of 0.16

Picking joint bandwidth of 0.178

Picking joint bandwidth of 0.147

Picking joint bandwidth of 0.136

Picking joint bandwidth of 0.15

Picking joint bandwidth of 0.133

Picking joint bandwidth of 0.188

Picking joint bandwidth of 0.123

Picking joint bandwidth of 0.17

Picking joint bandwidth of 0.143

Picking joint bandwidth of 0.138

Picking joint bandwidth of 0.182

Picking joint bandwidth of 0.158

Picking joint bandwidth of 0.151

Picking joint bandwidth of 0.101

Picking joint bandwidth of 0.134

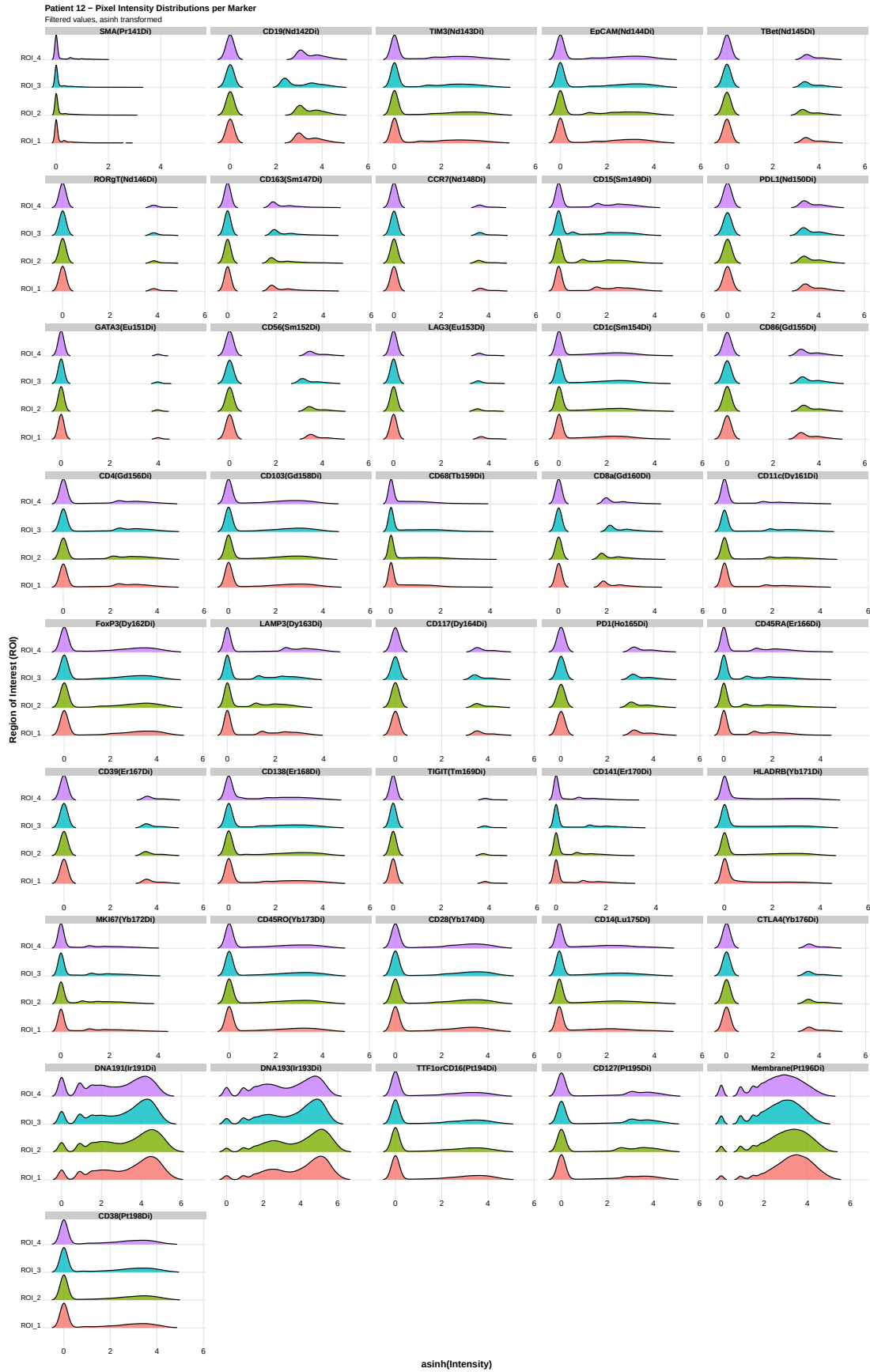
Picking joint bandwidth of 0.136

Picking joint bandwidth of 0.173

Picking joint bandwidth of 0.132

Picking joint bandwidth of 0.172

Picking joint bandwidth of 0.173
Picking joint bandwidth of 0.126
Picking joint bandwidth of 0.169
Picking joint bandwidth of 0.154
Picking joint bandwidth of 0.136
Picking joint bandwidth of 0.0939
Picking joint bandwidth of 0.137
Picking joint bandwidth of 0.12
Picking joint bandwidth of 0.161
Picking joint bandwidth of 0.171
Picking joint bandwidth of 0.143
Picking joint bandwidth of 0.17
Picking joint bandwidth of 0.157
Picking joint bandwidth of 0.158
Picking joint bandwidth of 0.172
Picking joint bandwidth of 0.179
Picking joint bandwidth of 0.102
Picking joint bandwidth of 0.167



No point in doing the asinh transformation on the filtered data because of the 0-100 scaling step.

Spatial and Distribution Visualization of Filtered Data

The spatial plots use raw intensity (not asinh) because the data is already 0-100 scaled.

Load ROI 1 Filtered Data (with X, Y coordinates)

```
roi1_filtered <- read.delim(  
  "/restricted/projectnb/camplab/home/vgupta7/tumor_imc/patient12_steinbock/raw_filtered/Patient_12T_ROI1_filtered.csv",  
  check.names = FALSE  
)  
  
cat("Dimensions:", nrow(roi1_filtered), "pixels x", ncol(roi1_filtered), "columns\n")  
  
## Dimensions: 1000000 pixels x 47 columns
```

Subset Filtered Data to Focused Markers

```
pixels_filtered_focused <- pixels_filtered %>%  
  filter(marker %in% focused_markers)  
  
cat("Focused markers confirmed:", length(unique(pixels_filtered_focused$marker)), "\n")  
  
## Focused markers confirmed: 16
```

Spatial Visualization: Selected Immune Markers (ROI 1, Filtered)

```
# Downsample to every 3rd pixel  
set.seed(42)  
roi1_filtered_sub <- roi1_filtered %>%  
  filter(X %% 3 == 0, Y %% 3 == 0)  
  
cat("Downsampled pixels:", nrow(roi1_filtered_sub), "\n")  
  
## Downsampled pixels: 111556  
  
# Same focused markers as before  
focused_markers_spatial <- c(  
  "EpCAM(Nd144Di)",  
  "SMA(Pr141Di)",  
  "FoxP3(Dy162Di)",  
  "CD68(Tb159Di)",  
  "CD163(Sm147Di)",  
  "CD1c(Sm154Di)",  
  "TIM3(Nd143Di)",  
  "PD1(Ho165Di)",  
  "TTF1orCD16(Pt194Di)"  
)  
  
roi1_filtered_long <- roi1_filtered_sub %>%  
  select(X, Y, all_of(focused_markers_spatial)) %>%  
  melt(id.vars = c("X", "Y"),  
       variable.name = "marker",
```

```

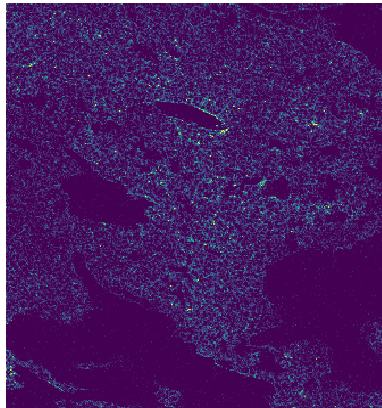
    value.name = "intensity")

p_filtered_spatial <- ggplot(roi1_filtered_long,
                             aes(x = X, y = Y, fill = intensity)) +
  geom_tile() +
  scale_fill_viridis_c(option = "viridis") +
  facet_wrap(~ marker, ncol = 3) +
  labs(
    title = "Patient 12 ROI 1 - Selected Immune Markers (Filtered)",
    subtitle = "Filtered values, 0-100 scaled, downsampled every 3rd pixel",
    fill = "Intensity\n(0-100)"
  ) +
  theme_void(base_size = 9) +
  theme(
    strip.text = element_text(size = 8, face = "bold"),
    legend.position = "bottom"
  )

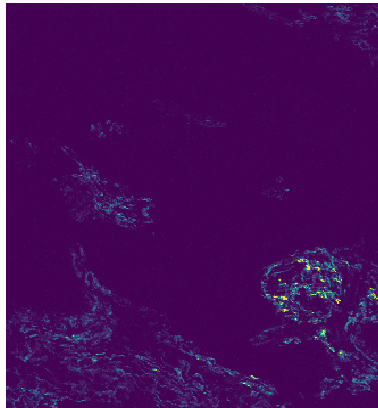
ggsave("Patient12T_ROI1_filtered_spatial.pdf", p_filtered_spatial, width = 12, height = 14)
print(p_filtered_spatial)

```

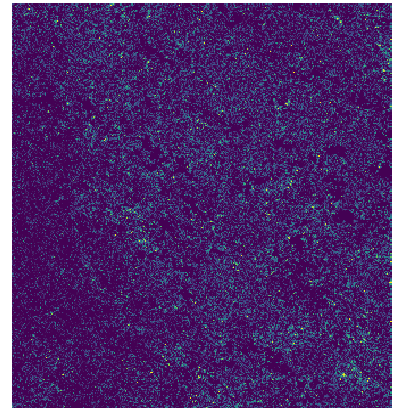
Patient 12 ROI 1 – Selected Immune Markers (Filtered)
 Filtered values, 0–100 scaled, downsampled every 3rd pixel
 EpCAM(Nd144Di)



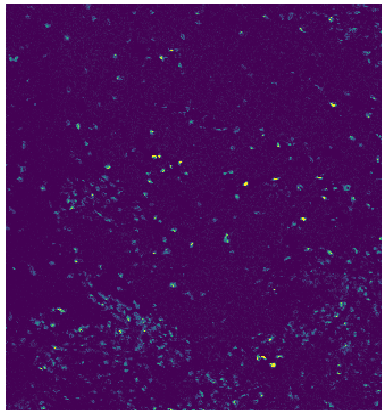
SMA(Pr141Di)



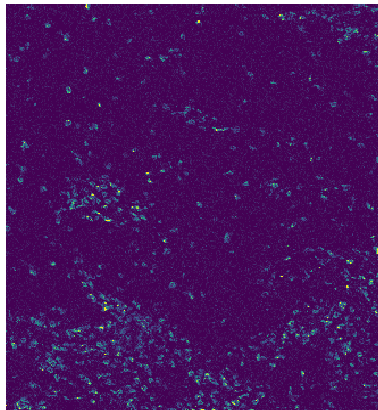
FoxP3(Dy162Di)



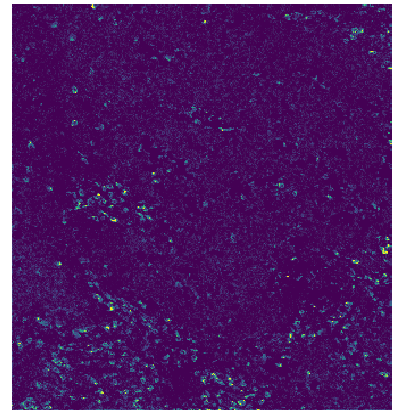
CD68(Tb159Di)



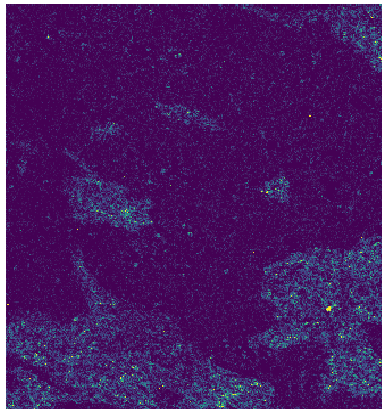
CD163(Sm147Di)



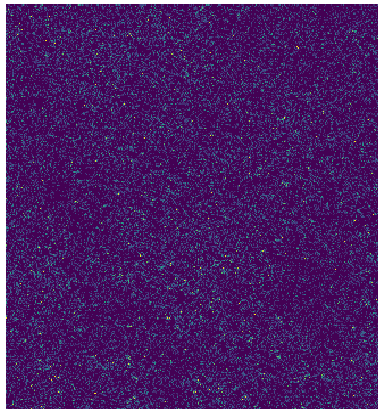
CD1c(Sm154Di)



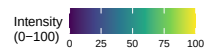
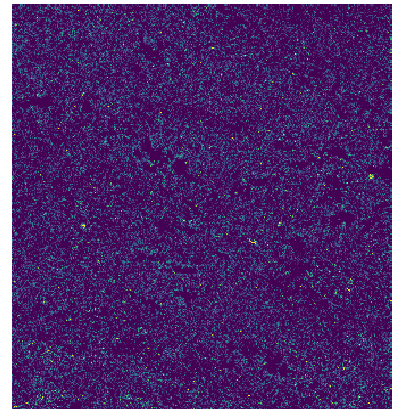
TIM3(Nd143Di)



PD1(Ho165Di)



TTF1orCD16(Pt194Di)



After filtering, there is a lot less background noise compared to the raw data spatial plots. Can now see discrete bright spots scattered across the tissue more clearly.

CD68 is a macrophage marker, so pixels with high CD68 expression are associated with macrophages.

But it's important to remember that we're looking at pixels, not cells, which is what cell segmentation is for.

FoxP3 and PD1 are rare/sparse populations that need cell-level aggregation after segmentation to detect reliably. There are some tiny yellow specks present, which means those cell types are present but rare in this

ROI.

Histograms: Focused Markers (ROI 1, Filtered)

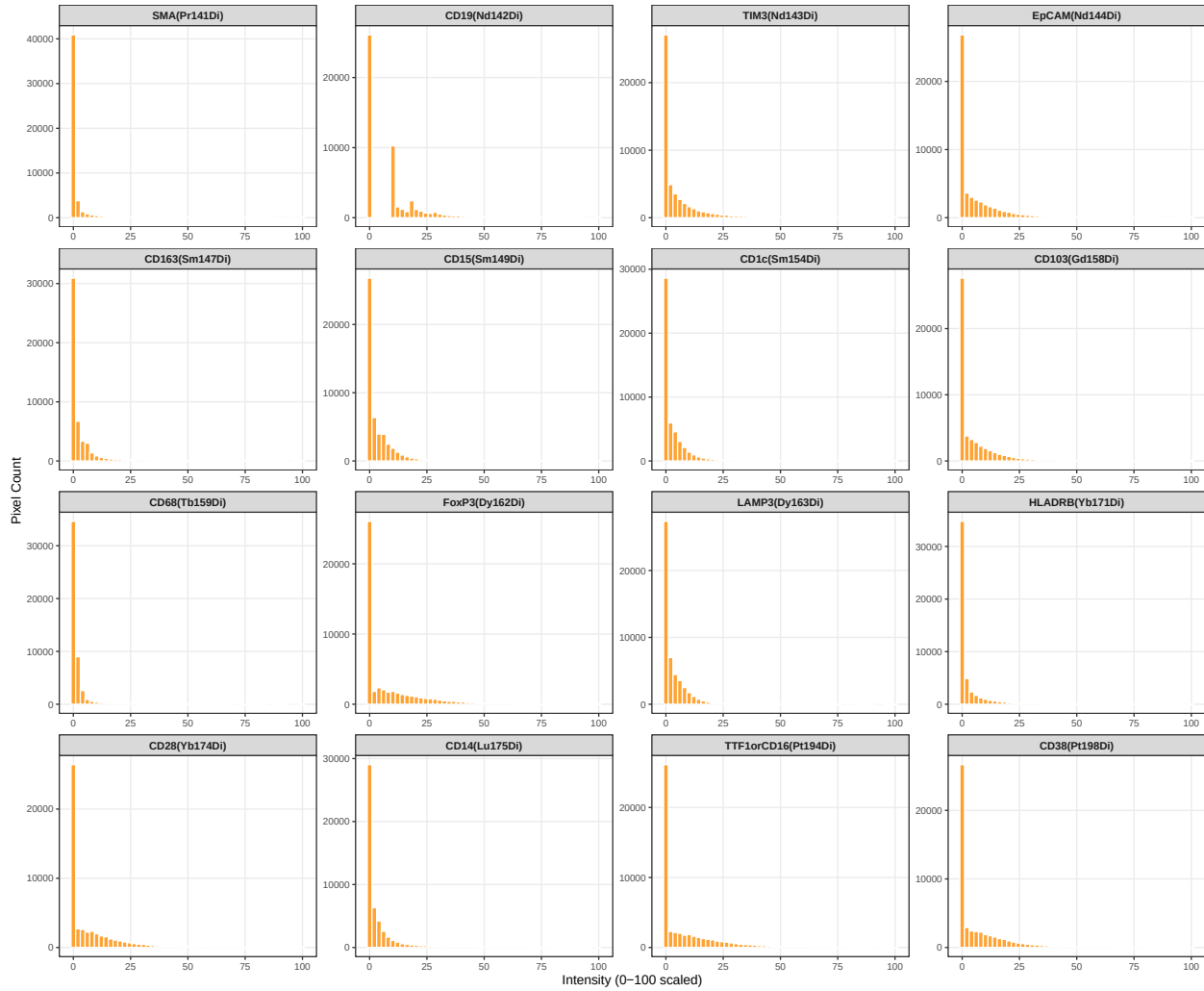
```
roi1_filtered_focused <- pixels_filtered_focused %>%
  filter(ROI == "ROI_1")

ggplot(roi1_filtered_focused, aes(x = intensity)) +
  geom_histogram(bins = 50, fill = "darkorange", color = "white",
                 size = 0.1, alpha = 0.8) +
  facet_wrap(~ marker, scales = "free", ncol = 4) +
  labs(
    title = "Pixel Intensity Landscape: Patient 12 (ROI 1, Filtered)",
    subtitle = "Filtered values, 0-100 scaled, 16 selected markers",
    x = "Intensity (0-100 scaled)",
    y = "Pixel Count"
  ) +
  theme_bw(base_size = 11) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(size = 8),
    axis.text.y = element_text(size = 8)
  )

## Warning in geom_histogram(bins = 50, fill = "darkorange", color = "white", :
## Ignoring unknown parameters: `size`
```

Pixel Intensity Landscape: Patient 12 (ROI 1, Filtered)

Filtered values, 0-100 scaled, 16 selected markers



Density Ridge Plot: Focused Markers (All ROIs, Filtered)

```
ggplot(pixels_filtered_focused, aes(x = intensity, y = ROI, fill = ROI)) +
  geom_density_ridges(alpha = 0.8, scale = 0.9, rel_min_height = 0.01,
    show.legend = FALSE) +
  facet_wrap(~ marker, scales = "free_x", ncol = 4) +
  labs(
    title = "Patient 12 - Pixel Intensity Distributions per Marker (Filtered)",
    subtitle = "Filtered values, 0-100 scaled, 16 selected markers",
    x = "Intensity (0-100 scaled)",
    y = "Region of Interest (ROI)"
  ) +
  theme_ridges(font_size = 10) +
  theme(
    strip.text = element_text(face = "bold", size = 9),
    axis.text.y = element_text(size = 8),
    axis.title = element_text(size = 11, face = "bold"),
    axis.title.x = element_text(hjust = 0.5),
    axis.title.y = element_text(hjust = 0.5, angle = 90)
```

)

```
## Picking joint bandwidth of 0.0543
## Picking joint bandwidth of 0.918
## Picking joint bandwidth of 0.562
## Picking joint bandwidth of 0.651
## Picking joint bandwidth of 0.247
## Picking joint bandwidth of 0.367
## Picking joint bandwidth of 0.364
## Picking joint bandwidth of 0.535
## Picking joint bandwidth of 0.136
## Picking joint bandwidth of 0.918
## Picking joint bandwidth of 0.415
## Picking joint bandwidth of 0.305
## Picking joint bandwidth of 0.872
## Picking joint bandwidth of 0.37
## Picking joint bandwidth of 0.879
## Picking joint bandwidth of 0.806
```

Patient 12 – Pixel Intensity Distributions per Marker (Filtered)

Filtered values, 0–100 scaled, 16 selected markers

